List DataStructure

In [1]:
```
l=[]
```

In [2]:
```
type(l)
```

Out[2]:
```
list
```

In [3]:
```
l.append(10)
l
```

Out[3]:
```
[10]
```

In [16]:
```
l.append(20)
l.append(30)
l.append(40)
l.append(50)
l.append(60)
l
```

Out[16]:
```
[10,
 10,
 10,
 10,
 10,
 10,
 20,
 30,
 40,
 50,
 10,
 20,
 30,
 40,
 50,
 10,
 20,
 30,
 40,
 50,
 10,
 20,
 20,
 30,
 40,
 50,
 60,
 20,
 30,
 40,
 50,
 60]
```

In [18]:
```
l.clear()
```

In [19]:
```
l
```

Out[19]:
```
[]
```

In [21]:
```
l.append(10)
l.append(20)
l.append(30)
l.append(40)
l.append(50)
l
```

Out[21]:
```
[10, 20, 30, 40, 50]
```

In [22]:
```
id(l)
```

Out[22]:
```
2762925802752
```

In [23]:
```
type(l)
```

Out[23]:
```
list
```

In [26]:
```
l.count(40)
```

Out[26]:
```
1
```

In [27]:
```
y=l.copy()
```

In [28]:
```
y
```

Out[28]:
```
[10, 20, 30, 40, 50]
```

In [29]:
```
len(y)
```

Out[29]:
```
5
```

In [30]:
```
len(l)
```

Out[30]:
```
5
```

In [31]:
```
l.clear()
```

In [32]:
```
l
```

Out[32]:
```
[]
```

In [33]:
```
y
```

Out[33]:
```
[10, 20, 30, 40, 50]
```

In [34]:
```
y.append('Ishwari')
```

In [35]:
```
y
```

Out[35]:
```
[10, 20, 30, 40, 50, 'Ishwari']
```

In [39]:
```
a=[2,4,5]
b=[7,9,6]
a.extend(b)
a
```

Out[39]:
```
[2, 4, 5, 7, 9, 6]
```

In [40]:
```
l.extend(y)
l
```

Out[40]:
```
[10, 20, 30, 40, 50, 'Ishwari']
```

In [41]:
```
y
```

Out[41]:
```
[10, 20, 30, 40, 50, 'Ishwari']
```

In [43]:
```
print(l)
print(y)
print(a)
```

```
[10, 20, 30, 40, 50, 'Ishwari']
[10, 20, 30, 40, 50, 'Ishwari']
[2, 4, 5, 7, 9, 6]
```

In [44]:
```
y[1]
```

Out[44]:
```
20
```

In [54]:
```
y[1]=100
```

In [55]:
```
y
```

Out[55]:
```
[10, 100, 30, 40, 50, 'Ishwari']
```

In [57]:
```
y.index(100) # if you want to check the index you want to put the value
```

Out[57]:
```
1
```

In [58]:
```
y
```

Out[58]:
```
[10, 100, 30, 40, 50, 'Ishwari']
```

In [59]:
```
y.insert(3,78)  # insert value after the 3rd index and value is 78
```

In [60]:
```
y
```

Out[60]:
```
[10, 100, 30, 78, 40, 50, 'Ishwari']
```

In [61]:
```
y.insert(6,65)
y
```

Out[61]:
```
[10, 100, 30, 78, 40, 50, 65, 'Ishwari']
```

In [62]:
```
y.insert(7,11)
y
```

Out[62]: [10, 100, 30, 78, 40, 50, 65, 11, 'Ishwari']

In [63]:
```
y.insert(9,55)
y
```

Out[63]: [10, 100, 30, 78, 40, 50, 65, 11, 'Ishwari', 55]

In [65]:
```
y.pop() # empty means eleminate last one by deafult
```

Out[65]: 55

In [66]:
```
y
```

Out[66]: [10, 100, 30, 78, 40, 50, 65, 11, 'Ishwari']

In [68]:
```
y.pop(3) # eliminated by indexing
```

Out[68]: 78

In [69]:
```
y
```

Out[69]: [10, 100, 30, 40, 50, 65, 11, 'Ishwari']

In [70]:
```
y.remove(11)  # remove by value
```

In [71]:
```
y
```

Out[71]: [10, 100, 30, 40, 50, 65, 'Ishwari']

In [75]:
```
y.pop()
```

Out[75]: 'Ishwari'

In [76]:
```
y
```

Out[76]: [10, 30, 40, 50, 65, 100]

In [77]:
```
y.append('Ishu')
```

In [78]:
```
y
```

Out[78]: [10, 30, 40, 50, 65, 100, 'Ishu']

In [79]:
```
y.sort()  # cuz in this list has string and also int
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[79], line 1
----> 1 y.sort()

TypeError: '<' not supported between instances of 'str' and 'int'
```

In [80]:
```
y.pop()
```

Out[80]: 'Ishu'

In [81]:
```
y
```
Out[81]:
```
[10, 30, 40, 50, 65, 100]
```
In [82]:
```
y.sort()
```
In [83]:
```
y
```
Out[83]:
```
[10, 30, 40, 50, 65, 100]
```
In [84]:
```
y.reverse()
```
In [85]:
```
y
```
Out[85]:
```
[100, 65, 50, 40, 30, 10]
```
In [86]:
```
y.append(98)
```
In [87]:
```
y
```
Out[87]:
```
[100, 65, 50, 40, 30, 10, 98]
```
In [89]:
```
y.sort(reverse=True)
```
In [90]:
```
y
```
Out[90]:
```
[100, 98, 65, 50, 40, 30, 10]
```
In [92]:
```
a=['Ishu', 'Isha', 'Ishwari']
a
```
Out[92]:
```
['Ishu', 'Isha', 'Ishwari']
```
In [93]:
```
a.sort()
```
In [94]:
```
a
```
Out[94]:
```
['Isha', 'Ishu', 'Ishwari']
```
In [95]:
```
a.reverse()
```
In [96]:
```
a
```
Out[96]:
```
['Ishwari', 'Ishu', 'Isha']
```
In [97]:
```
l3 = [10,20,[2,3],3]
```
In [106…
```
l3[2]=[8]
```
In [107…
```
l3
```
Out[107…
```
[10, 20, [8], 3]
```
In [108…
```
y
```

Out[108…
```
[100, 98, 65, 50, 40, 30, 10]
```

In [109…
```
50 in y
```

Out[109…
```
True
```

In [110…
```
200 in y
```

Out[110…
```
False
```

In [113…
```
for i in y:
    print(i)
```

```
100
98
65
50
40
30
10
```

In [ ]: